

Last updated: September 13, 2026

CONTACT INFORMATION	Email: jsung1@arizona.edu LinkedIn: linkedin.com/in/ejsung/ Citizenship: USA	Website: https://ejsung.github.io/ Location: Tucson, AZ
EDUCATION	University of Arizona, Tucson, AZ Doctor of Philosophy, Applied Mathematics Advisor: Dr. Charles A. Stafford Tulane University, New Orleans, LA Master of Science, Applied Mathematics Master's Thesis: <i>Mathematics and Applications of Feynman Diagrams</i> Advisor: Dr. Victor H. Moll Tulane University, New Orleans, LA Bachelor of Science, Mathematics Minor: Geology	Expected May 2028 May 2021 May 2020
PUBLICATIONS	[1] E. J. Sung , A. G. Campos, H. Abele, and D. I. Bondar, “Decoherence-free entropic gravity for a Dirac fermion”, <i>Physical Review D</i> 108 , 104036 (2023).	
UNDER REVIEW	[1] E. J. Sung and C. A. Stafford, <i>Decoherence and the Reemergence of Coherence From a Superconducting “Horizon”</i> , Mar. 2026, arXiv:2603.16765 [quant-ph] . [2] E. J. Sung , B. Koch, T. Jenke, H. Abele, and D. I. Bondar, <i>Generalized Boundary Conditions for the qBounce Experiment</i> , Oct. 2025, arXiv:2510.15341 [quant-ph] .	
PRESENTATIONS	[1] C. A. Stafford and E. J. Sung , “Decoherence from a superconducting “horizon””, in APS Meetings Archive , APS Global Physics Summit 2026, Abstract W32.00011 (American Physical Society, Mar. 2026). [2] J. Sung , B. Crooke, D. Carstens, C. Woock, R. Amer, and A. Kolker, “Sentinel-1A InSAR Time Series Analysis for Monitoring of Surface Subsidence in Coastal Louisiana”, in AGU Fall Meeting Abstracts, Vol. 2018 (Dec. 2018), G21C-0571.	
CONFERENCES AND WORKSHOPS	2026 MSTIC Summer Symposium Sandia National Laboratories, Albuquerque, NM - Quantum Thermodynamics & Decoherence (QTD) Symposium 2026 (Poster) University of Rochester, Rochester, NY - American Physical Society Global Physics Summit 2026 (Talk) Denver, CO - Math for All in New Orleans Tulane University, New Orleans, LA - Clifford Lectures Tulane University, New Orleans, LA - American Geophysical Union (Poster) Washington, DC	Aug 2026 Jun 2026 Mar 2026 Mar 2020 Oct 2019 Dec 2018
AWARDS AND HONORS	- Best Graduate Summer Student Presentation, Sandia National Laboratories - QTD Symposium Entropy Whisperer Poster Award (100 USD) - QTD Symposium Travel Award (1000 USD) Herbert E. Carter Travel Award (600 USD), Graduate College UArizona - Sigma Xi, The Scientific Research Honor Society, Full Member - University of Arizona Graduate College Fellowship	Aug 2026 Jun 2026 Jun 2026 Jan 2026 Dec 2025–Present Aug 2023
RESEARCH EXPERIENCE	Sandia National Laboratories, Albuquerque, NM <i>Year-Round Graduate Intern</i> <i>R&D Graduate Summer Intern</i> University of Arizona, Department of Physics, Tucson, AZ <i>Graduate Research Assistant</i> Developing a superconducting analog of black hole physics using non-equilibrium Green's functions (NEGFs) with Prof. Charles A. Stafford	August 2026–Present May 2026–August 2026 Dec 2024–Present

- Calculating NEGF transmission spectra for superconductor–normal devices (Landauer–Büttiker conductance) via simulations on Python and developing dephasing–probe extensions to estimate electronic phase-coherence time
- Implementing grid search algorithms on UA’s high-performance computing cluster to optimize superconducting parameters to increase transmission and phase coherence time
- Discovered solid-state analog of horizon-induced decoherence and novel reemergence of coherence
- [First-author paper](#) preprint submitted

University of Arizona, Department of Mathematics, Tucson, AZ

Graduate Research Assistant

May 2024–May 2025

- Implemented reinforcement learning algorithms using PyTorch to design optimal control strategies for particle accelerators under Prof. [Michael Chertkov](#)
- Developed a differentiable neural ODE surrogate of the accelerator beamline using training data generated with LANL’s HPSim, enabling rapid emulator-based testing of control ideas
- Designed and optimized particle accelerator simulations on UA’s high-performance computing cluster using CUDA for efficient parallel computations

Independent Studies Researcher

Sep 2023–Dec 2024

- Investigated quantum trajectories and decoherence of Schrödinger cat states under Prof. Janek Wehr
- Designed quantum Monte Carlo simulations of Schrödinger cat states in optical cavities using QuTiP on Python (see [GitHub repo](#))
- Worked on formulating Weierstrass–Enneper representations of the Dirac equation with Prof. Patrick Shipman, discovered a general symmetry theorem

Army Educational Outreach Program Undergraduate Internship, Tulane University, Department of Physics, New Orleans, LA

Research Intern

May 2023–Aug 2023

- Collaborated with Prof. [Denys I. Bondar](#) and the Institut Laue–Langevin to investigate standard gravity discrepancies of free-falling neutrons observed in the 2023 qBounce experiment
- Discovered, via open quantum systems simulations using QuTiP on Python and Mathematica, that the experimental boundary conditions were responsible for the discrepancy
- [First-author paper](#) preprint submitted

Tulane University, Department of Physics, New Orleans, LA

Research Assistant

May 2021–Aug 2023

- Conducted research with Prof. [Denys I. Bondar](#) on entropic gravitational effects on Dirac fermions in an open quantum systems framework
- Formulated a Lindblad master equation for free-falling Dirac fermions, demonstrating coherence under strong gravitational coupling and aligning with experimental results
- Discovered a novel zitterbewegung gravitational effect via open quantum systems simulations using Python and Mathematica; published findings in [first-author paper](#) in Physical Review D

Tulane University, Department of Mathematics, New Orleans, LA

Master’s Research

Aug 2020–May 2021

- Conducted master’s thesis research, under Prof. [Victor H. Moll](#), on the Feynman diagrams and quantum field theory (QFT) along with their applications
- Developed proficiency in the application of QFT, scattering theory, electrodynamics, and Lie theory
- Successfully defended thesis to thesis committee and submitted written [master’s thesis](#) to Tulane University Thesis & Dissertations Archive

Tulane University, Department of Physics, New Orleans, LA

Research Assistant

May 2020–Aug 2020

- Investigated the accuracy and applications of the Equations of Motion (EOM) method under Prof. [George Rosensteel](#)

- Designed and implemented EOM method using Python to the Morse potential, calculating excitation energies in higher dimensions with high precision
- Gained proficiency in Python and computational physics

Tulane University, Department of Mathematics, New Orleans, LA

Senior Seminar Research

Aug 2019–May 2020

- Conducted year-long research under Prof. [Lisa J. Fauci](#) investigating the collective phase of *Bacillus subtilis* bacteria in viscous flows
- Implemented regularization algorithms on Python and MATLAB to compute forces and velocities of bacterial motion to determine the origin of collective phase
- Completed an [expository paper](#) and gave a presentation of results to Senior Seminar class

Tulane University, Department of Earth & Environmental Sciences, New Orleans, LA

Research Assistant

Jun 2018–Dec 2018

- Investigated land subsidence in Golden Meadows, Coastal Louisiana, under Prof. [Reda Amer](#)
- Collected and analyzed over 100 Real-Time Kinematic GPS elevation points and Sentinel-1A satellite InSAR remote sensing data
- Implemented interpolation and image classification algorithms on ArcGIS and ENVI, discovering a 2.91% land reduction (90 km²) from 1985-2017 at 0.2-1.64 cm/year
- Presented findings as a [first-author poster](#) at the Fall 2018 American Geophysical Union in Washington D.C.

PROJECTS

Predicting Grant Success: A Multimethod Approach Using TDA, NLP, and Bibliometric Analysis

[Paper](#), [GitHub Repo](#)

Oct 2024–Dec 2024

- Developed and implemented machine learning models (Logistic Regression, Random Forest, and Gradient Boosting) using R and Python's scikit-learn to predict Department of Defense grant approval rates based on bibliometric data
- Enhanced prediction accuracy by fine-tuning hyperparameters for Random Forest and Gradient Boosting models, achieving a Test AUC-ROC of 0.743 and improving overall classification accuracy to 71.3%
- Analyzed feature importance and identified i10-index and total citations as the most influential predictors, providing actionable insights into grant success factors for academic researchers

Numerical Solutions to Reaction Diffusion Equations

[Paper](#)

Oct 2020–Dec 2020

- Investigated long-time dynamics and pattern formation in 1D/2D reaction-diffusion PDEs under varied boundary conditions
- Built MATLAB finite-difference solvers using Crank–Nicolson (diffusion), ADI (2D), and explicit multistep updates for nonlinear reaction terms
- Implemented Dirichlet, Neumann (ghost-node), and periodic BCs with efficient tridiagonal/LU linear solves to ensure stable, scalable time-stepping
- Reproduced Gray–Scott two-species pattern regimes; generated heatmaps/cross-sections and documented parameter sensitivity and emergent structures in a formal report

TEACHING

University of Arizona, Department of Mathematics, Tucson, AZ

Graduate Teaching Assistant

Aug 2023–Present

- Solo instructor for Math 112 College Algebra
- Graduate teaching assistant for Math 186J College Algebra Supplemental Instruction Seminar, Math 196V Vector Calculus Supplemental Instruction Seminar, and Math 468 Applied Stochastic Processes
- Hosted weekly office hours, graded homework assignments, and proctored exams

Walter L. Cohen High School, New Orleans, LA

Tutor

Aug 2016–Dec 2016

- Taught math, general sciences, English, and history at Walter L. Cohen High School
- Collaborated with other teachers in designing and implementing new teaching methods and problem sets

LEADERSHIP AND OUTREACH	Tulane University Earth Science Club, New Orleans, LA <i>President</i> - Established and led a student earth science club, growing membership to 40 students - Secured up to \$1000 in funding through successful grant proposals to the undergraduate student government for supporting club activities - Organized and coordinated events, including hiking trips, camping excursions, and networking workshops	May 2018–Dec 2018
SKILLS	Programming/Software: Python (pandas, NumPy, SciPy, SymPy, Scikit-learn, TensorFlow, PyTorch, QuTiP), Julia, R, Mathematica, CUDA, MATLAB, C++, SQL, HTML, CSS, JavaScript, Git, Microsoft Office, ArcGIS, ENVI, Jupyter, LaTeX Languages: Korean (Elementary), Spanish (Elementary)	
PROFESSIONAL SOCIETIES	American Physical Society (APS) Society for Industrial and Applied Mathematics (SIAM) American Mathematical Society (AMS)	Aug 2024–Present Jan 2022–Present Jan 2024–Present